

直流电桥与电阻测量

李谨硕, IEEE 试点班, 电院 2504

2026 年 6 月 1 日

1 实验原理

1.1 惠斯登电桥法

惠斯登电桥将两个已知比例臂电阻、一个电阻箱和待测电阻组成桥式网络。调节电阻箱 R_0 , 使检流计 G 中电流为零, 此时桥路达到平衡, 桥臂中点电势相等, 待测电阻可由比例关系求得。

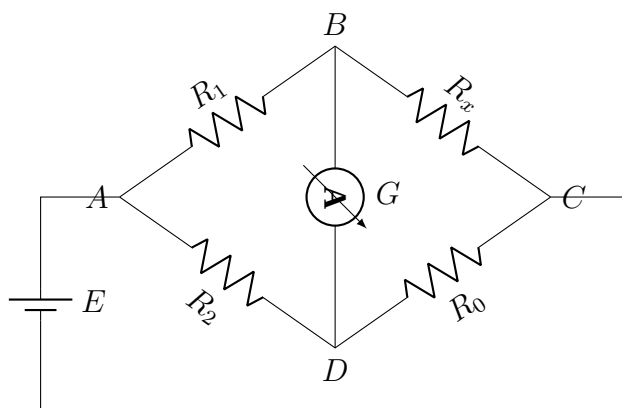


图 1: 惠斯登电桥原理图。平衡时检流计无偏转, $U_B = U_D$ 。

平衡条件为

$$\frac{R_1}{R_2} = \frac{R_x}{R_0}, \quad R_x = \frac{R_1}{R_2} R_0. \quad (1)$$

其中 R_1 、 R_2 为标准电阻, R_0 为电阻箱读数。电桥灵敏度按给定数据定义为

$$S = \frac{\Delta U_0}{\Delta R_0 / R_0}. \quad (2)$$

式中 ΔR_0 是平衡点附近给定的电阻箱扰动量, ΔU_0 是相应的电压变化量。由于原始数据中 ΔU_0 的单位为 mV, 本文中 S 的单位也记为 mV。

对电阻箱读数 R_0 , 三次测量平均值、样本标准差和 A 类不确定度为

$$\bar{R}_0 = \frac{1}{n} \sum_{i=1}^n R_{0i}, \quad \sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (R_{0i} - \bar{R}_0)^2}, \quad \Delta_A = 2.48\sigma. \quad (3)$$

这里 $n = 3$, 系数 2.48 来自 PPT 中给出的 $t_{0.95}/\sqrt{n}$ 。B 类不确定度由电阻箱各档读数逐位按相对等级加权求和, 并计入残余电阻 0.020Ω :

$$\Delta_B = \sum_j R_j \gamma_j + 0.020\Omega, \quad (4)$$

其中 R_j 是对应档位实际贡献的阻值, γ_j 是该档位相对不确定度。合成不确定度为

$$u(R_0) = \sqrt{\Delta_A^2 + \Delta_B^2}. \quad (5)$$

标准电阻 R_1 、 R_2 的相对不确定度均取 0.05%, 则

$$\frac{u(R_x)}{R_x} = \sqrt{\left(\frac{u(R_1)}{R_1}\right)^2 + \left(\frac{u(R_2)}{R_2}\right)^2 + \left(\frac{u(R_0)}{R_0}\right)^2}. \quad (6)$$

对第二组数据的交换法补充实验, 取两次平衡读数 R_0 与 R'_0 的几何平均得到最终电阻值, 并按 PPT 中公式计算相对不确定度:

$$R_x = \sqrt{R_0 R'_0}, \quad \frac{u(R_x)}{R_x} = \frac{1}{2} \sqrt{\left(\frac{u(R_0)}{R_0}\right)^2 + \left(\frac{u(R'_0)}{R'_0}\right)^2}. \quad (7)$$

1.2 四端法

四端法用于低阻测量。外侧两端通入电流, 内侧两端测量电压。电压表内阻很大, 内侧电压端几乎无电流流过, 因此电压端引线和接触电阻造成的压降不计入测量电压, 能够显著减小低阻测量误差。

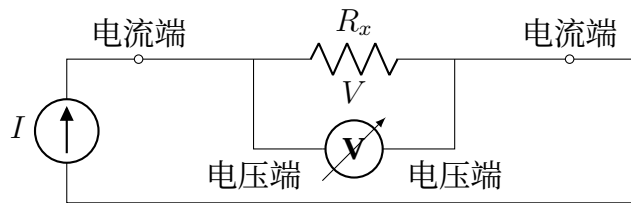


图 2: 四端法测量低电阻的接线原理图。

对每一组 U - I 数据作一次函数最小二乘拟合:

$$U = kI + b, \quad (8)$$

其中 U 用 mV 表示, I 用 mA 表示, 因此斜率 k 的单位为 $\text{mV}/\text{mA} = \Omega$, 即被测电阻。若被测低阻试样可视为圆柱形导体, 长度为 L , 直径为 d , 横截面积为

$$A = \frac{\pi d^2}{4}, \quad (9)$$

则四端法测得的电阻率为

$$\rho = \frac{RA}{L} = \frac{k\pi d^2}{4L}. \quad (10)$$

本报告中 d 与 L 的原始单位为 mm，计算电阻率时统一换算为 SI 单位。等价地，若直接代入 d_{mm} 、 L_{mm} ，则

$$\rho/(\Omega \cdot \text{m}) = k \cdot \frac{\pi d_{\text{mm}}^2}{4L_{\text{mm}}} \times 10^{-3}. \quad (11)$$

最小二乘估计式为

$$k = \frac{\sum_i (I_i - \bar{I})(U_i - \bar{U})}{\sum_i (I_i - \bar{I})^2}, \quad b = \bar{U} - k\bar{I}. \quad (12)$$

拟合优度用

$$R^2 = 1 - \frac{\sum_i (U_i - kI_i - b)^2}{\sum_i (U_i - \bar{U})^2} \quad (13)$$

R^2 越接近 1，线性关系越好。

2 实验目的

1. 掌握惠斯登电桥平衡测电阻的基本原理，理解比例臂选择、电桥平衡和电桥灵敏度的物理意义。
2. 学会根据多次平衡读数计算待测电阻，并按 A 类、B 类不确定度公式完成不确定度评定。
3. 掌握四端法测量低电阻的接线思想，理解其消除引线电阻和接触电阻影响的原因，并由几何尺寸计算试样电阻率。
4. 使用 Python 对实验数据进行自动化计算、线性最小二乘拟合和图像输出，提高数据处理的可复现性。

3 数据记录表

3.1 惠斯登电桥法原始数据

表 1 中 R_x 为万用表粗测值，用于选择比例臂；1、2、3 为三次电阻箱平衡读数。

表 1: 惠斯登电桥法原始数据

编号	粗测 R_x/Ω	R_1/Ω	R_2/Ω	1/ Ω	2/ Ω	3/ Ω	I/mA	$\Delta R_0/\Omega$	$\Delta U_0/\text{mV}$
A	1.102e+05	10000	1000	11027.3	11027.2	11027.1	0.875	1	0.06
B	913.4	1000	1000	912.7	912.5	912.8	10.982	1	0.01
C	50.82	100	10000	5077.0	5076.7	5077.4	7.262	1	0.04

表 2: 第二组交换法补充实验原始数据

编号	粗测 R_x/Ω	R_1/Ω	R_2/Ω	$1/\Omega$	$2/\Omega$	$3/\Omega$	I/mA	$\Delta R_0/\Omega$	$\Delta U_0/\text{mV}$
B'	913.3	1000	1000	912.8	912.6	912.3	11.020	1	0.01

3.2 四端法原始数据

表 3: 四端法 U - I 原始数据

U_1/mV	I_1/mA	U_2/mV	I_2/mA	U_3/mV	I_3/mA
1.10	14.93	0.45	21.13	2.92	28.10
1.60	33.65	0.81	38.31	3.84	37.00
2.24	47.40	1.24	58.16	4.98	48.00
2.91	61.76	1.55	72.72	6.80	65.51
3.87	82.00	1.92	92.36	8.61	82.94
4.68	99.12	2.37	111.97	9.84	94.70
5.37	113.56	2.70	126.89	11.55	111.21
6.07	128.61	3.22	151.42	12.78	123.09
6.59	139.60	3.66	171.87	14.41	138.68
7.40	156.53	3.91	183.80	16.08	154.73

表 4: 四端法试样几何尺寸

组号	直径 d/mm	有效长度 L/mm	说明
1	0.506	14.20	与第 1 组 U_1 - I_1 数据对应
2	0.510	14.20	与第 2 组 U_2 - I_2 数据对应
3	0.452	14.20	与第 3 组 U_3 - I_3 数据对应

4 实验方案及仪器

4.1 实验方案

本实验分为两部分。第一部分采用惠斯登电桥法测量三个待测电阻，先用万用表获得粗测值，再根据粗测值选择合适的比例臂 R_1 、 R_2 ，调节电阻箱 R_0 到检流计无偏转，重

复三次记录平衡读数。随后在平衡点附近改变 ΔR_0 ，记录相应 ΔU_0 ，用于计算电桥灵敏度。第二组数据另做一次交换法补充实验，用于比较常规电桥法与交换法的不确定度。

第二部分采用四端法测量三组低电阻。实验中改变电流并记录相应电压，得到三组 $U-I$ 数据。用 Python 对每组数据分别拟合 $U = kI + b$ ，取斜率作为被测电阻，再结合试样直径 d 与有效长度 L 计算电阻率 $\rho = RA/L$ 。

4.2 主要仪器

根据实验数据和测量方法，本实验使用的仪器与器材包括：直流稳压电源、检流计或微伏计、标准电阻或比例臂电阻 R_1 与 R_2 、十进制电阻箱 R_0 、待测电阻、数字万用表、四端法低阻测量接线装置、恒流源、电压表以及连接导线。

5 实验过程及现象

5.1 惠斯登电桥法

首先用万用表粗测待测电阻，确定其数量级。根据粗测值选择 R_1 与 R_2 ，使电阻箱平衡读数落在合适范围内。接通电源后，缓慢调节 R_0 ，观察检流计偏转方向。当 R_0 偏离平衡点时，检流计向一侧偏转；调节方向反向后，检流计偏转方向随之改变。逐步缩小调节步长后，检流计示数趋近零，说明桥路两中点电势相等。每个电阻记录三次平衡读数，以降低随机误差。

在灵敏度测量中，在平衡点附近人为改变 ΔR_0 。当电桥灵敏度较高时，较小的相对电阻扰动即可产生可观察的电压变化；当灵敏度较低时，同样的扰动对应的电压变化更小，平衡点判定更容易受仪器分辨率影响。

5.2 四端法

四端法实验中，电流由外侧电流端通过待测低阻，电压由内侧电压端测量。调节电流后，三组数据的电压均随电流近似线性增大。拟合图中数据点基本落在直线附近，说明在实验电流范围内待测元件近似满足欧姆定律，没有明显的发热非线性或接触不稳定现象。

6 实验数据处理及结果

6.1 惠斯登电桥法结果

按第 1 节公式计算三次平衡读数平均值、A 类不确定度、B 类不确定度、合成不确定度和待测电阻，结果见表 5。表中相对不确定度以百分数表示。为保证计算过程可复核，下面给出全部代入过程。电阻箱 B 类不确定度按有效读数所在档位逐位求和，最后统一加残余电阻 0.020Ω 。

第一组 A:

$$\bar{R}_{0,A} = \frac{11027.3 + 11027.2 + 11027.1}{3} = 11027.2000\Omega,$$

$$\sigma_A = \sqrt{\frac{(11027.3 - 11027.2)^2 + (11027.2 - 11027.2)^2 + (11027.1 - 11027.2)^2}{2}} = 0.1000\Omega,$$

$$\Delta_{A,A} = 2.48 \times 0.1000 = 0.2480\Omega,$$

$$\Delta_{B,A} = 10000 \times 0.001 + 1000 \times 0.001 + 20 \times 0.001 \\ + 7 \times 0.005 + 0.2 \times 0.020 + 0.020 = 11.0790\Omega,$$

$$u(R_{0,A}) = \sqrt{0.2480^2 + 11.0790^2} = 11.0818\Omega,$$

$$R_A = \frac{10000}{1000} \times 11027.2000 = 110272.0000\Omega,$$

$$u_r(R_A) = \sqrt{0.0005^2 + 0.0005^2 + \left(\frac{11.0818}{11027.2000}\right)^2} = 0.0012288,$$

$$u(R_A) = 110272.0000 \times 0.0012288 = 135.5010\Omega,$$

$$S_A = \frac{0.06}{1/11027.2000} = 661.6320\text{mV}.$$

第二组 B:

$$\bar{R}_{0,B} = \frac{912.7 + 912.5 + 912.8}{3} = 912.6667\Omega,$$

$$\sigma_B = \sqrt{\frac{(912.7 - 912.6667)^2 + (912.5 - 912.6667)^2 + (912.8 - 912.6667)^2}{2}} = 0.1528\Omega,$$

$$\Delta_{A,B} = 2.48 \times 0.1528 = 0.3788\Omega,$$

$$\Delta_{B,B} = 900 \times 0.001 + 10 \times 0.001 + 2 \times 0.005 + 0.6667 \times 0.020 + 0.020 = 0.9533\Omega,$$

$$u(R_{0,B}) = \sqrt{0.3788^2 + 0.9533^2} = 1.0258\Omega,$$

$$R_B = \frac{1000}{1000} \times 912.6667 = 912.6667\Omega,$$

$$u_r(R_B) = \sqrt{0.0005^2 + 0.0005^2 + \left(\frac{1.0258}{912.6667}\right)^2} = 0.0013279,$$

$$u(R_B) = 912.6667 \times 0.0013279 = 1.2120\Omega,$$

$$S_B = \frac{0.01}{1/912.6667} = 9.1267\text{mV}.$$

第三组 C:

$$\bar{R}_{0,C} = \frac{5077.0 + 5076.7 + 5077.4}{3} = 5077.0333\Omega,$$

$$\sigma_C = \sqrt{\frac{(5077.0 - 5077.0333)^2 + (5076.7 - 5077.0333)^2 + (5077.4 - 5077.0333)^2}{2}} = 0.3512\Omega,$$

$$\Delta_{A,C} = 2.48 \times 0.3512 = 0.8709\Omega,$$

$$\Delta_{B,C} = 5000 \times 0.001 + 70 \times 0.001 + 7 \times 0.005 + 0.0333 \times 0.020 + 0.020 = 5.1257\Omega,$$

$$u(R_{0,C}) = \sqrt{0.8709^2 + 5.1257^2} = 5.1991\Omega,$$

$$R_C = \frac{100}{10000} \times 5077.0333 = 50.7703\Omega,$$

$$u_r(R_C) = \sqrt{0.0005^2 + 0.0005^2 + \left(\frac{5.1991}{5077.0333}\right)^2} = 0.0012445,$$

$$u(R_C) = 50.7703 \times 0.0012445 = 0.0632\Omega,$$

$$S_C = \frac{0.04}{1/5077.0333} = 203.0813\text{mV}.$$

表 5: 惠斯登电桥法数据处理结果

编号	$\bar{R}_0/\Omega$	R_x/Ω	Δ_A/Ω	Δ_B/Ω	$u(R_0)/\Omega$	$u_r(R_x)/\%$	$u(R_x)/\Omega$	S/mV
A	11027.2000	110272.0000	0.2480	11.0790	11.0818	0.1229	135.5010	661.6320
B	912.6667	912.6667	0.3788	0.9533	1.0258	0.1328	1.2120	9.1267
C	5077.0333	50.7703	0.8709	5.1257	5.1991	0.1244	0.0632	203.0813

由此得到三个待测电阻的测量结果为

$$R_A = (110272 \pm 136) \Omega,$$

$$R_B = (912.67 \pm 1.21) \Omega,$$

$$R_C = (50.770 \pm 0.063) \Omega.$$

上述不确定度均按 README 给定公式计算。第一组电阻较大, 绝对不确定度较大, 但相对不确定度约为 0.123%; 第三组电阻值最小, 绝对不确定度最小, 但相对不确定度仍与其他组同量级。

交换法补充实验结果见表 6。交换法利用两个平衡读数的几何平均, 避免单独引入比例臂标准电阻不确定度, 因此第二组电阻的最终不确定度由 1.21Ω 降至 0.77Ω。其代入计

算为

$$\begin{aligned}\bar{R}'_0 &= \frac{912.8 + 912.6 + 912.3}{3} = 912.5667\Omega, \\ \sigma' &= \sqrt{\frac{(912.8 - 912.5667)^2 + (912.6 - 912.5667)^2 + (912.3 - 912.5667)^2}{2}} = 0.2517\Omega, \\ \Delta'_A &= 2.48 \times 0.2517 = 0.6241\Omega, \\ \Delta'_B &= 900 \times 0.001 + 10 \times 0.001 + 2 \times 0.005 + 0.5667 \times 0.020 + 0.020 = 0.9513\Omega, \\ u(R'_0) &= \sqrt{0.6241^2 + 0.9513^2} = 1.1378\Omega, \\ R_B^{\text{ex}} &= \sqrt{912.6667 \times 912.5667} = 912.6167\Omega, \\ u_r(R_B^{\text{ex}}) &= \frac{1}{2} \sqrt{\left(\frac{1.0258}{912.6667}\right)^2 + \left(\frac{1.1378}{912.5667}\right)^2} = 0.0008393, \\ u(R_B^{\text{ex}}) &= 912.6167 \times 0.0008393 = 0.7660\Omega.\end{aligned}$$

表 6: 第二组交换法补充实验数据处理结果

编号	$\bar{R}'_0/\Omega$	Δ'_A/Ω	Δ'_B/Ω	$u(R'_0)/\Omega$	$u_r(R'_0)/\%$	$\sqrt{R_0 R'_0}/\Omega$	$u(R_x)/\Omega$
B'	912.5667	0.6241	0.9513	1.1378	0.1247	912.6167	0.7660

交换法给出的第二组电阻结果为

$$R_B = (912.62 \pm 0.77) \Omega.$$

该结果与常规电桥法的 912.67Ω 一致, 说明实验重复性良好。

6.2 四端法线性拟合、电阻和电阻率结果

四端法数据由 `plot.py` 进行最小二乘拟合, 拟合图见图 3。由 `calc.py` 计算拟合斜率、斜率标准误差、试样横截面积和电阻率。

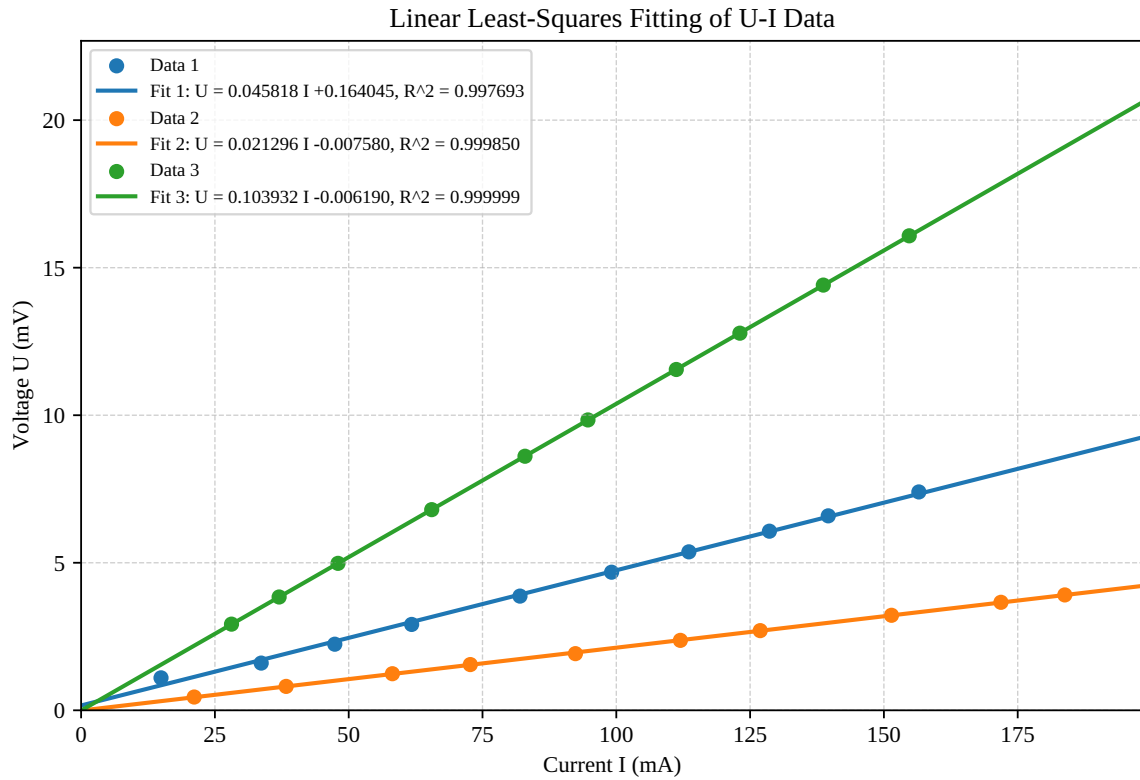


图 3: 四端法 U - I 数据的一次函数最小二乘拟合。

最小二乘计算采用

$$k = \frac{n \sum IU - \sum I \sum U}{n \sum I^2 - (\sum I)^2}, \quad b = \frac{\sum U - k \sum I}{n}.$$

三组数据的求和量见表 7。

表 7: 四端法最小二乘拟合求和量

组号	n	$\sum I/\text{mA}$	$\sum U/\text{mV}$	$\sum IU/\text{mA mV}$	$\sum I^2/\text{mA}^2$
1	10	877.16	41.83	4606.1481	97391.2660
2	10	1028.63	21.83	2845.9517	134003.3929
3	10	883.96	91.81	9898.6043	95293.6312

逐组代入如下。第一组:

$$k_1 = \frac{10 \times 4606.1481 - 877.16 \times 41.83}{10 \times 97391.2660 - 877.16^2} = 0.0458178\Omega,$$

$$b_1 = \frac{41.83 - 0.0458178 \times 877.16}{10} = 0.164045\text{mV},$$

$$A_1 = \frac{\pi(0.506\text{mm})^2}{4} = 0.201090\text{mm}^2,$$

$$\rho_1 = 0.0458178 \times \frac{0.201090}{14.20} \times 10^{-3} = 6.48839 \times 10^{-7}\Omega \cdot \text{m} = 0.648839 \mu\Omega \cdot \text{m}.$$

第二组:

$$\begin{aligned}k_2 &= \frac{10 \times 2845.9517 - 1028.63 \times 21.83}{10 \times 134003.3929 - 1028.63^2} = 0.0212961\Omega, \\b_2 &= \frac{21.83 - 0.0212961 \times 1028.63}{10} = -0.007580\text{mV}, \\A_2 &= \frac{\pi(0.510\text{mm})^2}{4} = 0.204282\text{mm}^2, \\\rho_2 &= 0.0212961 \times \frac{0.204282}{14.20} \times 10^{-3} = 3.06367 \times 10^{-7}\Omega \cdot \text{m} = 0.306367 \mu\Omega \cdot \text{m}.\end{aligned}$$

第三组:

$$\begin{aligned}k_3 &= \frac{10 \times 9898.6043 - 883.96 \times 91.81}{10 \times 95293.6312 - 883.96^2} = 0.1039322\Omega, \\b_3 &= \frac{91.81 - 0.1039322 \times 883.96}{10} = -0.006190\text{mV}, \\A_3 &= \frac{\pi(0.452\text{mm})^2}{4} = 0.160460\text{mm}^2, \\\rho_3 &= 0.1039322 \times \frac{0.160460}{14.20} \times 10^{-3} = 1.17443 \times 10^{-6}\Omega \cdot \text{m} = 1.174434 \mu\Omega \cdot \text{m}.\end{aligned}$$

拟合参数见表 8。由于 U 的单位为 mV、 I 的单位为 mA，拟合斜率数值即为电阻值，单位为 Ω 。

表 8: 四端法一次函数拟合结果

组号	斜率 k/Ω	截距 b/mV	R^2	斜率标准误差/ Ω
1	0.045818	0.164045	0.997693	0.000779
2	0.021296	-0.007580	0.999850	0.000092
3	0.103932	-0.006190	0.999999	0.000034

结合试样几何尺寸得到的电阻率见表 9。表中电阻率标准误差仅由拟合斜率标准误差传播得到，未计入直径 d 与长度 L 的仪器允差；原因是题目只给出 d 、 L 的读数，没有给出其测量不确定度。

表 9: 四端法电阻率计算结果

组号	d/mm	L/mm	A/mm^2	R/Ω	$\rho/(\Omega \cdot \text{m})$	$\rho/(\mu\Omega \cdot \text{m})$	$u_\rho/(\mu\Omega \cdot \text{m})$
1	0.506	14.20	0.201090	0.045818	6.48839×10^{-7}	0.648839	0.011032
2	0.510	14.20	0.204282	0.021296	3.06367×10^{-7}	0.306367	0.001327
3	0.452	14.20	0.160460	0.103932	1.17443×10^{-6}	1.174434	0.000383

因此四端法得到的最终电阻和电阻率为

$$\begin{aligned} R_1^{(4T)} &= 0.045818\Omega, & \rho_1 &= 6.48839 \times 10^{-7}\Omega \cdot \text{m}, \\ R_2^{(4T)} &= 0.021296\Omega, & \rho_2 &= 3.06367 \times 10^{-7}\Omega \cdot \text{m}, \\ R_3^{(4T)} &= 0.103932\Omega, & \rho_3 &= 1.17443 \times 10^{-6}\Omega \cdot \text{m}. \end{aligned}$$

三组拟合的 R^2 分别为 0.997693、0.999850 和 0.999999，均接近 1，说明四端法数据具有很好的线性关系。其中第一组的截距为 0.164045mV，相对更明显，可能来自零点偏置、接触状态或低电流端读数误差；第二、三组截距接近零，线性关系更理想。

7 实验总结及讨论

本实验完成了中高阻的惠斯登电桥测量和低阻的四端法测量。惠斯登电桥法的核心在于调节 R_0 使检流计无偏转，从而把待测电阻转换为比例臂与电阻箱读数的乘积。实验中三组电桥测量结果分别为 $(110272 \pm 136)\Omega$ 、 $(912.67 \pm 1.21)\Omega$ 和 $(50.770 \pm 0.063)\Omega$ 。补充交换法对第二组电阻得到 $(912.62 \pm 0.77)\Omega$ ，与常规法一致，且不确定度更小。

误差来源主要包括以下几类：第一，电桥平衡点判定依赖检流计灵敏度和操作者调节步长，平衡位置附近的微小偏差会直接影响 R_0 读数；第二，电阻箱不同档位的相对不确定度不同，高阻档对第一组测量的不确定度贡献较大；第三，标准电阻 R_1 、 R_2 的等级误差会通过比例关系传递到 R_x ；第四，温度变化可能改变待测电阻和标准电阻的实际阻值。

四端法部分中，三组 U - I 曲线均表现出良好线性，拟合斜率分别为 0.045818Ω 、 0.021296Ω 和 0.103932Ω 。结合 $d = 0.506\text{mm}$ 、 0.510mm 、 0.452mm 以及 $L = 14.20\text{mm}$ 的几何数据，三组电阻率分别为 $6.48839 \times 10^{-7}\Omega \cdot \text{m}$ 、 $3.06367 \times 10^{-7}\Omega \cdot \text{m}$ 和 $1.17443 \times 10^{-6}\Omega \cdot \text{m}$ 。四端法能有效排除电流引线和接触电阻的压降，特别适合毫欧到零点几欧量级的低阻测量。若采用普通二端法，引线电阻和接触电阻可能与待测低阻同量级，从而造成显著系统误差。

从数据处理角度看，使用 Python 脚本统一完成读数平均、A/B 类不确定度、交换法计算、线性拟合、电阻率计算和作图，可以避免手算取位不一致，保证实验报告中的数据与源数据可追溯。后续若更换数据，只需更新 data.csv、脚本中的电桥读数或四端法几何尺寸，即可重新生成全部结果。

8 应用延伸与拓展

惠斯登电桥不仅可用于普通电阻测量，也广泛用于传感器电路。例如应变片、热敏电阻和铂电阻温度传感器常被接入桥式电路，通过桥路不平衡电压反映微小电阻变化。由于电桥可把微小阻值变化转换为电压信号，它适合与放大器和数据采集系统结合，实现应力、温度、压力等物理量的精密测量。

四端法也称 Kelvin 测量法，是低阻测量和大电流器件测试中的标准方法。它常用于测量电缆电阻、PCB 走线电阻、焊点接触电阻、分流器电阻、电池内阻以及功率器件导通电阻。进一步拓展时，可使用恒流源和纳伏表组成更高精度的四端测量系统；也可采用交流四端法减小热电势和直流漂移对微小电压测量的影响。

本实验还体现了实验自动化处理的重要性。对大量 U - I 数据或多组电桥读数，脚本化处理可直接生成结果文件、图像和表格，便于重复实验、误差复核和报告归档。

9 附件

9.1 附件 A：计算脚本 calc.py

```

1  #!/usr/bin/env python3
2  """Compute all numerical results for the DC bridge and four-terminal experiment.
3
4  Run this script in the same directory as data.csv:
5
6      python calc.py
7
8  The script writes calc_results.json and LaTeX tables under ./tables/.
9  """
10
11  from __future__ import annotations
12
13  import json
14  import math
15  from pathlib import Path
16
17  import numpy as np
18  import pandas as pd
19
20  T_FACTOR_OVER_SQRT_N = 2.48
21  STANDARD_RESISTOR_REL_UNCERTAINTY = 0.0005
22  RESIDUAL_RESISTANCE_OHM = 0.020
23
24  DECADE_RELATIVE_UNCERTAINTY = {
25      10000.0: 0.001,
26      1000.0: 0.001,
27      100.0: 0.001,
28      10.0: 0.001,
29      1.0: 0.005,
30      0.1: 0.020,
31  }
32
33  WHEATSTONE_DATA = [
34      {
35          "label": "A",
36          "rough_Rx_ohm": 1.1022e5,
37          "R1_ohm": 1.0e4,
38          "R2_ohm": 1.0e3,
39          "R0_readings_ohm": [11027.3, 11027.2, 11027.1],
40          "I_mA": 0.875,
41          "delta_R0_ohm": 1.0,
42          "delta_U0_mV": 0.06,
43      },
44      {
45          "label": "B",
46          "rough_Rx_ohm": 913.4,
47          "R1_ohm": 1.0e3,
48          "R2_ohm": 1.0e3,
49          "R0_readings_ohm": [912.7, 912.5, 912.8],
50          "I_mA": 10.982,
51          "delta_R0_ohm": 1.0,
52          "delta_U0_mV": 0.01,

```

```

53     },
54     {
55         "label": "C",
56         "rough_Rx_ohm": 50.82,
57         "R1_ohm": 1.0e2,
58         "R2_ohm": 1.0e4,
59         "R0_readings_ohm": [5077.0, 5076.7, 5077.4],
60         "I_mA": 7.262,
61         "delta_R0_ohm": 1.0,
62         "delta_U0_mV": 0.04,
63     },
64 ]
65
66 EXCHANGE_DATA = {
67     "label": "B",
68     "rough_Rx_ohm": 913.3,
69     "R1_ohm": 1.0e3,
70     "R2_ohm": 1.0e3,
71     "R0_readings_ohm": [912.8, 912.6, 912.3],
72     "I_mA": 11.020,
73     "delta_R0_ohm": 1.0,
74     "delta_U0_mV": 0.01,
75 }
76
77 FOUR_TERMINAL_DIMENSIONS = [
78     {"index": 1, "diameter_mm": 0.506, "length_mm": 14.20},
79     {"index": 2, "diameter_mm": 0.510, "length_mm": 14.20},
80     {"index": 3, "diameter_mm": 0.452, "length_mm": 14.20},
81 ]
82
83
84 def fmt(x: float, digits: int = 4) -> str:
85     """Format a floating-point number for LaTeX tables."""
86     return f"{x:.{digits}f}"
87
88
89 def decade_b_uncertainty(resistance_ohm: float) -> tuple[float, list[dict[str, float]]]:
90     """Return B-type uncertainty of the resistance box by decade summation."""
91     remainder = float(resistance_ohm)
92     total = 0.0
93     components: list[dict[str, float]] = []
94
95     for decade in [10000.0, 1000.0, 100.0, 10.0, 1.0, 0.1]:
96         if decade >= 1.0:
97             digit = math.floor(remainder / decade + 1e-12)
98             component_value = digit * decade
99             remainder -= component_value
100         else:
101             component_value = max(remainder, 0.0)
102             remainder = 0.0
103
104     contribution = component_value * DECADE_RELATIVE_UNCERTAINTY[decade]
105     components.append(
106         {
107             "decade_ohm": decade,
108             "component_ohm": component_value,
109             "relative_uncertainty": DECADE_RELATIVE_UNCERTAINTY[decade],

```

```

110         "contribution_ohm": contribution,
111     }
112 )
113 total += contribution
114
115 total += RESIDUAL_RESISTANCE_OHM
116 return total, components
117
118
119 def process_bridge_record(record: dict[str, object], include_standard_resistors: bool = True) -> dict[str,
    object]:
120     """Compute bridge value, sensitivity, and uncertainty for one record."""
121     readings = np.array(record["R0_readings_ohm"], dtype=float)
122     R0_mean = float(np.mean(readings))
123     sample_sigma = float(np.std(readings, ddof=1))
124     delta_A = T_FACTOR_OVER_SQRT_N * sample_sigma
125     delta_B, b_components = decade_b_uncertainty(R0_mean)
126     u_R0 = math.sqrt(delta_A**2 + delta_B**2)
127     u_rel_R0 = u_R0 / R0_mean
128
129     ratio = float(record["R1_ohm"]) / float(record["R2_ohm"])
130     Rx = ratio * R0_mean
131
132     if include_standard_resistors:
133         u_rel_Rx = math.sqrt(
134             STANDARD_RESISTOR_REL_UNCERTAINTY**2
135             + STANDARD_RESISTOR_REL_UNCERTAINTY**2
136             + u_rel_R0**2
137         )
138     else:
139         u_rel_Rx = u_rel_R0
140
141     u_Rx = Rx * u_rel_Rx
142     sensitivity_mV = float(record["delta_U0_mV"]) / (float(record["delta_R0_ohm"]) / R0_mean)
143
144     return {
145         **record,
146         "R0_mean_ohm": R0_mean,
147         "sample_sigma_ohm": sample_sigma,
148         "delta_A_ohm": delta_A,
149         "delta_B_ohm": delta_B,
150         "u_R0_ohm": u_R0,
151         "u_rel_R0": u_rel_R0,
152         "ratio_R1_over_R2": ratio,
153         "Rx_ohm": Rx,
154         "u_rel_Rx": u_rel_Rx,
155         "u_Rx_ohm": u_Rx,
156         "sensitivity_mV": sensitivity_mV,
157         "B_components": b_components,
158     }
159
160
161 def linear_fit_with_errors(current_mA: np.ndarray, voltage_mV: np.ndarray) -> dict[str, float]:
162     """Fit  $U = kI + b$  and return  $k$ ,  $b$ ,  $R^2$ , and standard errors."""
163     slope, intercept = np.polyfit(current_mA, voltage_mV, deg=1)
164     fitted = slope * current_mA + intercept
165     residual = voltage_mV - fitted

```

```

166     ss_res = float(np.sum(residual**2))
167     ss_tot = float(np.sum((voltage_mV - np.mean(voltage_mV)) ** 2))
168     r_squared = 1.0 - ss_res / ss_tot if ss_tot != 0 else float("nan")
169
170     n = len(current_mA)
171     dof = n - 2
172     residual_variance = ss_res / dof
173     sxx = float(np.sum((current_mA - np.mean(current_mA)) ** 2))
174     slope_stderr = math.sqrt(residual_variance / sxx)
175     intercept_stderr = math.sqrt(
176         residual_variance * (1.0 / n + float(np.mean(current_mA)) ** 2 / sxx)
177     )
178
179     return {
180         "n": int(n),
181         "sum_I_mA": float(np.sum(current_mA)),
182         "sum_U_mV": float(np.sum(voltage_mV)),
183         "sum_IU_mA_mV": float(np.sum(current_mA * voltage_mV)),
184         "sum_I2_mA2": float(np.sum(current_mA * current_mA)),
185         "sum_U2_mV2": float(np.sum(voltage_mV * voltage_mV)),
186         "slope_ohm": float(slope),
187         "intercept_mV": float(intercept),
188         "r_squared": float(r_squared),
189         "slope_stderr_ohm": float(slope_stderr),
190         "intercept_stderr_mV": float(intercept_stderr),
191     }
192
193
194 def add_resistivity_results(fit_results: list[dict[str, float]]) -> None:
195     """Add wire cross-sectional area and resistivity to each four-terminal fit."""
196     dimensions = {item["index"]: item for item in FOUR_TERMINAL_DIMENSIONS}
197     for fit in fit_results:
198         dim = dimensions[fit["index"]]
199         diameter_mm = float(dim["diameter_mm"])
200         length_mm = float(dim["length_mm"])
201         area_mm2 = math.pi * diameter_mm**2 / 4.0
202         resistivity_ohm_m = fit["slope_ohm"] * area_mm2 / length_mm * 1.0e-3
203         resistivity_stderr_ohm_m = fit["slope_stderr_ohm"] * area_mm2 / length_mm * 1.0e-3
204         fit.update(
205             {
206                 "diameter_mm": diameter_mm,
207                 "length_mm": length_mm,
208                 "area_mm2": area_mm2,
209                 "resistivity_ohm_m": resistivity_ohm_m,
210                 "resistivity_micro_ohm_m": resistivity_ohm_m * 1.0e6,
211                 "resistivity_stderr_ohm_m": resistivity_stderr_ohm_m,
212                 "resistivity_stderr_micro_ohm_m": resistivity_stderr_ohm_m * 1.0e6,
213             }
214         )
215
216
217 def write_latex_tables(results: dict[str, object], out_dir: Path) -> None:
218     """Write LaTeX table fragments used by report.tex."""
219     out_dir.mkdir(parents=True, exist_ok=True)
220
221     bridge_rows = []
222     for item in results["wheatstone"]:

```

```

223     r = item["R0_readings_ohm"]
224     bridge_rows.append(
225         f"{item['label']}& {item['rough_Rx_ohm']:.4g} & {item['R1_ohm']:.0f} & {item['R2_ohm']:.0f} & "
226         f"{r[0]:.1f} & {r[1]:.1f} & {r[2]:.1f} & {item['I_mA']:.3f} & "
227         f"{item['delta_R0_ohm']:.0f} & {item['delta_U0_mV']:.2f} \\\\"
228     )
229     (out_dir / "wheatstone_raw_rows.tex").write_text("\n".join(bridge_rows), encoding="utf-8")
230
231     ex = results["exchange"]
232     r = ex["R0_readings_ohm"]
233     exchange_row = (
234         f"{ex['label']}& {ex['rough_Rx_ohm']:.4g} & {ex['R1_ohm']:.0f} & {ex['R2_ohm']:.0f} & "
235         f"{r[0]:.1f} & {r[1]:.1f} & {r[2]:.1f} & {ex['I_mA']:.3f} & "
236         f"{ex['delta_R0_ohm']:.0f} & {ex['delta_U0_mV']:.2f} \\\\"
237     )
238     (out_dir / "exchange_raw_row.tex").write_text(exchange_row, encoding="utf-8")
239
240     processed_rows = []
241     for item in results["wheatstone"]:
242         processed_rows.append(
243             f"{item['label']}& {item['R0_mean_ohm']:.4f} & {item['Rx_ohm']:.4f} & "
244             f"{item['delta_A_ohm']:.4f} & {item['delta_B_ohm']:.4f} & {item['u_R0_ohm']:.4f} & "
245             f"{100*item['u_rel_Rx']:.4f} & {item['u_Rx_ohm']:.4f} & {item['sensitivity_mV']:.4f} \\\\"
246         )
247     (out_dir / "wheatstone_processed_rows.tex").write_text("\n".join(processed_rows), encoding="utf-8")
248
249     exchange_result = results["exchange_result"]
250     exchange_processed_row = (
251         f"{ex['label']}& {ex['R0_mean_ohm']:.4f} & {ex['delta_A_ohm']:.4f} & "
252         f"{ex['delta_B_ohm']:.4f} & {ex['u_R0_ohm']:.4f} & {100*ex['u_rel_R0']:.4f} & "
253         f"{exchange_result['Rx_exchange_ohm']:.4f} & {exchange_result['u_Rx_exchange_ohm']:.4f} \\\\"
254     )
255     (out_dir / "exchange_processed_row.tex").write_text(exchange_processed_row, encoding="utf-8")
256
257     fit_rows = []
258     for fit in results["four_terminal_fits"]:
259         fit_rows.append(
260             f"{fit['index']}& {fit['slope_ohm']:.6f} & {fit['intercept_mV']:.6f} & "
261             f"{fit['r_squared']:.6f} & {fit['slope_stderr_ohm']:.6f} \\\\"
262         )
263     (out_dir / "four_terminal_fit_rows.tex").write_text("\n".join(fit_rows), encoding="utf-8")
264
265     resistivity_rows = []
266     for fit in results["four_terminal_fits"]:
267         resistivity_rows.append(
268             f"{fit['index']}& {fit['diameter_mm']:.3f} & {fit['length_mm']:.2f} & "
269             f"{fit['area_mm2']:.6f} & {fit['slope_ohm']:.6f} & "
270             f"{fit['resistivity_ohm_m']:.6e} & {fit['resistivity_micro_ohm_m']:.6f} \\"
271         )
272     (out_dir / "four_terminal_resistivity_rows.tex").write_text("\n".join(resistivity_rows), encoding="utf-8")
273
274     data = pd.read_csv(Path(__file__).resolve().parent / "data.csv")
275     data.columns = [c.strip() for c in data.columns]
276     data_rows = []
277     for _, row in data.iterrows():

```

```

278     data_rows.append(
279         f"{row['U1(mV)']:.2f} & {row['I1(mA)']:.2f} & "
280         f"{row['U2(mV)']:.2f} & {row['I2(mA)']:.2f} & "
281         f"{row['U3(mV)']:.2f} & {row['I3(mA)']:.2f} \\\\"
282     )
283     (out_dir / "four_terminal_raw_rows.tex").write_text("\n".join(data_rows), encoding="utf-8")
284
285
286 def main() -> None:
287     base_dir = Path(__file__).resolve().parent
288     data_path = base_dir / "data.csv"
289     if not data_path.exists():
290         raise FileNotFoundError(f"Cannot find {data_path}")
291
292     bridge_results = [process_bridge_record(record) for record in WHEATSTONE_DATA]
293     exchange_result = process_bridge_record(EXCHANGE_DATA, include_standard_resistors=False)
294
295     R0 = bridge_results[1]["R0_mean_ohm"]
296     R0_exchange = exchange_result["R0_mean_ohm"]
297     Rx_exchange = math.sqrt(R0 * R0_exchange)
298     u_rel_exchange = 0.5 * math.sqrt(
299         bridge_results[1]["u_rel_R0"] ** 2 + exchange_result["u_rel_R0"] ** 2
300     )
301     u_Rx_exchange = Rx_exchange * u_rel_exchange
302
303     data = pd.read_csv(data_path)
304     data.columns = [column.strip() for column in data.columns]
305     fit_results = []
306     for index in range(1, 4):
307         current = data[f"I{index}(mA)"].to_numpy(dtype=float)
308         voltage = data[f"U{index}(mV)"].to_numpy(dtype=float)
309         fit_results.append({"index": index, **linear_fit_with_errors(current, voltage)})
310     add_resistivity_results(fit_results)
311
312     results = {
313         "constants": {
314             "T_FACTOR_OVER_SQRT_N": T_FACTOR_OVER_SQRT_N,
315             "STANDARD_RESISTOR_REL_UNCERTAINTY": STANDARD_RESISTOR_REL_UNCERTAINTY,
316             "RESIDUAL_RESISTANCE_OHM": RESIDUAL_RESISTANCE_OHM,
317             "DECADE_RELATIVE_UNCERTAINTY": DECADE_RELATIVE_UNCERTAINTY,
318             "FOUR_TERMINAL_DIMENSIONS": FOUR_TERMINAL_DIMENSIONS,
319         },
320         "wheatstone": bridge_results,
321         "exchange": exchange_result,
322         "exchange_result": {
323             "Rx_exchange_ohm": Rx_exchange,
324             "u_rel_Rx_exchange": u_rel_exchange,
325             "u_Rx_exchange_ohm": u_Rx_exchange,
326         },
327         "four_terminal_fits": fit_results,
328     }
329
330     with (base_dir / "calc_results.json").open("w", encoding="utf-8") as f:
331         json.dump(results, f, indent=2, ensure_ascii=False)
332
333     write_latex_tables(results, base_dir / "tables")
334

```

```
335 print("Wheatstone bridge results:")
336 for item in bridge_results:
337     print(
338         f" {item['label']}: Rx = {item['Rx_ohm']:.6g} ohm, "
339         f"u(Rx) = {item['u_Rx_ohm']:.4g} ohm, "
340         f"S = {item['sensitivity_mV']:.4g} mV"
341     )
342 print(
343     "Exchange method: "
344     f"Rx = {Rx_exchange:.6g} ohm, u(Rx) = {u_Rx_exchange:.4g} ohm"
345 )
346 print("Four-terminal fits:")
347 for fit in fit_results:
348     print(
349         f" {fit['index']}: U = {fit['slope_ohm']:.6f} I "
350         f"{fit['intercept_mV']:+.6f}, R^2 = {fit['r_squared']:.6f}, "
351         f"rho = {fit['resistivity_ohm_m']:.6e} ohm*m"
352     )
353
354
355 if __name__ == "__main__":
356     main()
```

9.2 附件 B: 作图脚本 plot.py

```

1  #!/usr/bin/env python3
2  """Plot and fit the U-I data for the four-terminal resistance experiment.
3
4  Run this script in the same directory as data.csv:
5
6      python plot.py
7
8  The script reads three U-I column pairs, fits each pair with
9
10      $U = \text{slope} * I + \text{intercept}$ 
11
12  and saves a single PDF figure named ui_fit.pdf.
13  """
14
15  from pathlib import Path
16
17  import matplotlib
18  matplotlib.use("Agg")
19  import matplotlib.pyplot as plt
20  from matplotlib import font_manager
21  import numpy as np
22  import pandas as pd
23
24
25  CSV_NAME = "data.csv"
26  PDF_NAME = "ui_fit.pdf"
27
28
29  def configure_font() -> str:
30      """Configure a Times New Roman style font for Matplotlib."""
31      available_fonts = {font.name for font in font_manager.fontManager.ttflist}
32
33      for font_name in ["Times New Roman", "Tinos", "Liberation Serif", "DejaVu Serif"]:
34          if font_name in available_fonts:
35              plt.rcParams["font.family"] = font_name
36              break
37      else:
38          font_name = "serif"
39          plt.rcParams["font.family"] = "serif"
40
41      plt.rcParams["pdf.fonttype"] = 42
42      plt.rcParams["ps.fonttype"] = 42
43      plt.rcParams["mathtext.fontset"] = "stix"
44      return font_name
45
46
47  def linear_fit(x: np.ndarray, y: np.ndarray) -> tuple[float, float, float]:
48      """Return slope, intercept, and R^2 for the fit  $y = \text{slope} * x + \text{intercept}$ ."""
49      slope, intercept = np.polyfit(x, y, deg=1)
50      y_pred = slope * x + intercept
51      ss_res = np.sum((y - y_pred) ** 2)
52      ss_tot = np.sum((y - np.mean(y)) ** 2)
53      r_squared = 1.0 - ss_res / ss_tot if ss_tot != 0 else np.nan
54      return slope, intercept, r_squared
55

```

```

56
57 def axis_start_point(slope: float, intercept: float) -> tuple[float, float]:
58     """Return the visible positive-axis intersection of  $y = \text{slope} \cdot x + \text{intercept}$ ."""
59     if intercept >= 0:
60         return 0.0, intercept
61     if slope > 0:
62         return -intercept / slope, 0.0
63     return 0.0, max(intercept, 0.0)
64
65
66 def main() -> None:
67     """Read data.csv, fit three U-I pairs, and save the PDF plot."""
68     configure_font()
69
70     script_dir = Path(__file__).resolve().parent
71     csv_path = script_dir / CSV_NAME
72     pdf_path = script_dir / PDF_NAME
73
74     if not csv_path.exists():
75         raise FileNotFoundError(f"Cannot find {CSV_NAME} in {script_dir}")
76
77     data = pd.read_csv(csv_path)
78     data.columns = [column.strip() for column in data.columns]
79
80     fig, ax = plt.subplots(figsize=(7.2, 5.0))
81     fit_results = []
82     all_x_values = []
83     all_y_values = []
84
85     for index in range(1, 4):
86         u_col = f"U{index}(mV)"
87         i_col = f"I{index}(mA)"
88         if u_col not in data.columns or i_col not in data.columns:
89             raise KeyError(f"Missing required columns: {u_col} and/or {i_col}")
90
91         pair_data = data[[i_col, u_col]].dropna()
92         x = pair_data[i_col].to_numpy(dtype=float)
93         y = pair_data[u_col].to_numpy(dtype=float)
94         slope, intercept, r_squared = linear_fit(x, y)
95         x_start, y_start = axis_start_point(slope, intercept)
96
97         fit_results.append({
98             "index": index,
99             "x": x,
100             "y": y,
101             "slope": slope,
102             "intercept": intercept,
103             "r_squared": r_squared,
104             "x_start": x_start,
105             "y_start": y_start,
106         })
107         all_x_values.extend(x.tolist())
108         all_y_values.extend(y.tolist())
109
110     x_max = max(all_x_values) * 1.08
111     for result in fit_results:
112         y_end = result["slope"] * x_max + result["intercept"]

```

```
113     all_y_values.extend([result["y_start"], y_end])
114     y_max = max(all_y_values) * 1.10
115
116     for result in fit_results:
117         index = result["index"]
118         slope = result["slope"]
119         intercept = result["intercept"]
120         r_squared = result["r_squared"]
121
122         ax.scatter(result["x"], result["y"], s=32, marker="o", label=f"Data {index}")
123         x_line = np.linspace(result["x_start"], x_max, 300)
124         y_line = slope * x_line + intercept
125         ax.plot(
126             x_line,
127             y_line,
128             linewidth=1.8,
129             label=f"Fit {index}: U = {slope:.6f} I {intercept:+.6f}, R^2 = {r_squared:.6f}",
130         )
131
132         print(
133             f"Fit {index}: U = {slope:.6f} I {intercept:+.6f}; "
134             f"R^2 = {r_squared:.6f}"
135         )
136
137     ax.set_xlim(left=0, right=x_max)
138     ax.set_ylim(bottom=0, top=y_max)
139     ax.set_xlabel("Current I (mA)")
140     ax.set_ylabel("Voltage U (mV)")
141     ax.set_title("Linear Least-Squares Fitting of U-I Data")
142     ax.grid(True, linestyle="--", linewidth=0.6, alpha=0.6)
143     ax.legend(fontsize=8.2, frameon=True)
144     fig.tight_layout()
145     fig.savefig(pdf_path, bbox_inches="tight")
146     plt.close(fig)
147
148     print(f"Saved PDF: {pdf_path}")
149
150
151 if __name__ == "__main__":
152     main()
```